Confidence Intervals

Lecture 10

Topics in this lecture

- Central Limit Theorem
- Large-sample confidence interval for a population mean

Central Limit Theorem

- If we draw a large enough sample from the population, then the distribution of the *sample mean* is approximately *normal*, no matter what population the sample was drawn from

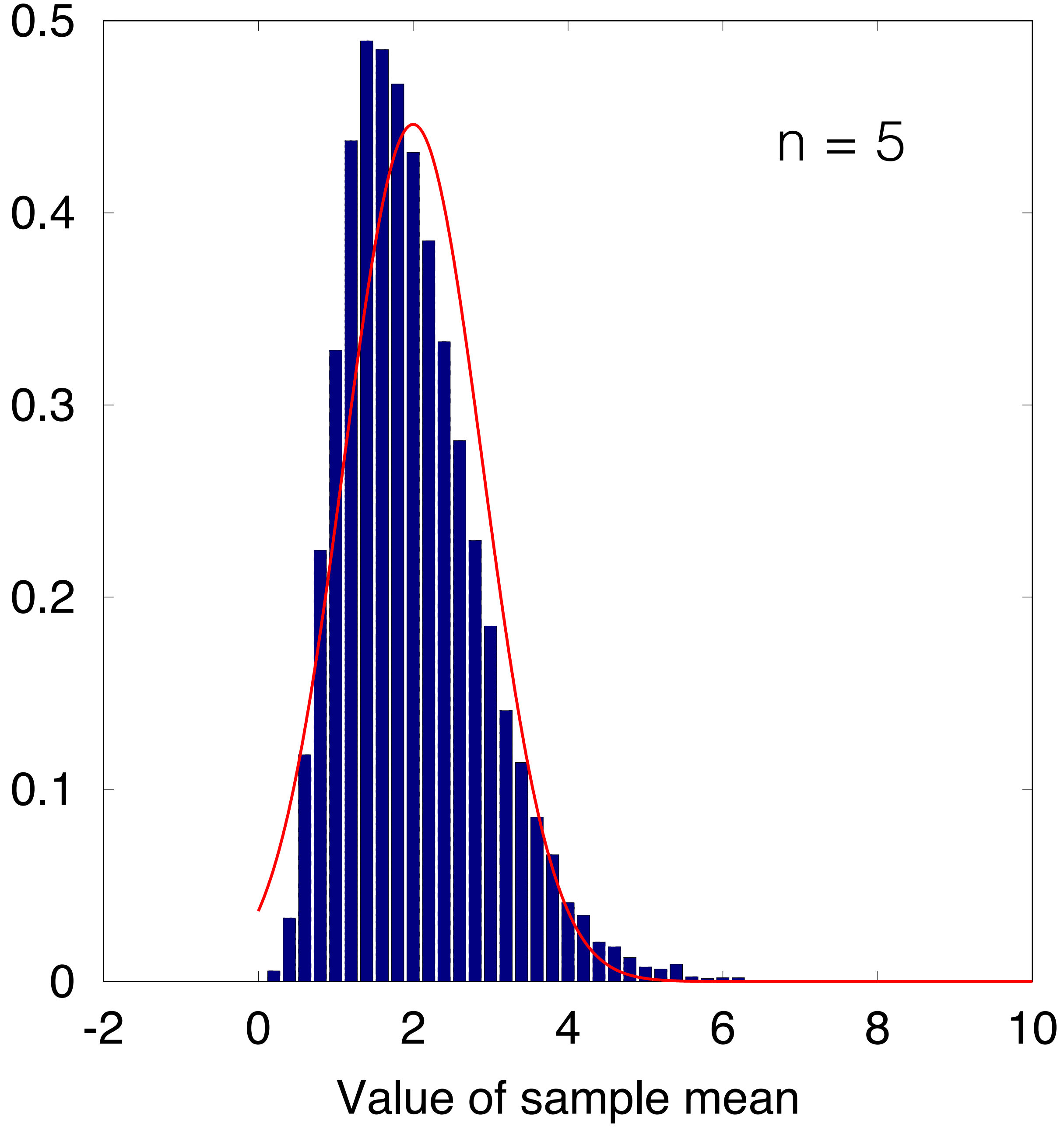
- Let $X_1, X_2, \ldots, X_n$ be a simple random sample drawn from a population with mean μ and variance σ^2,

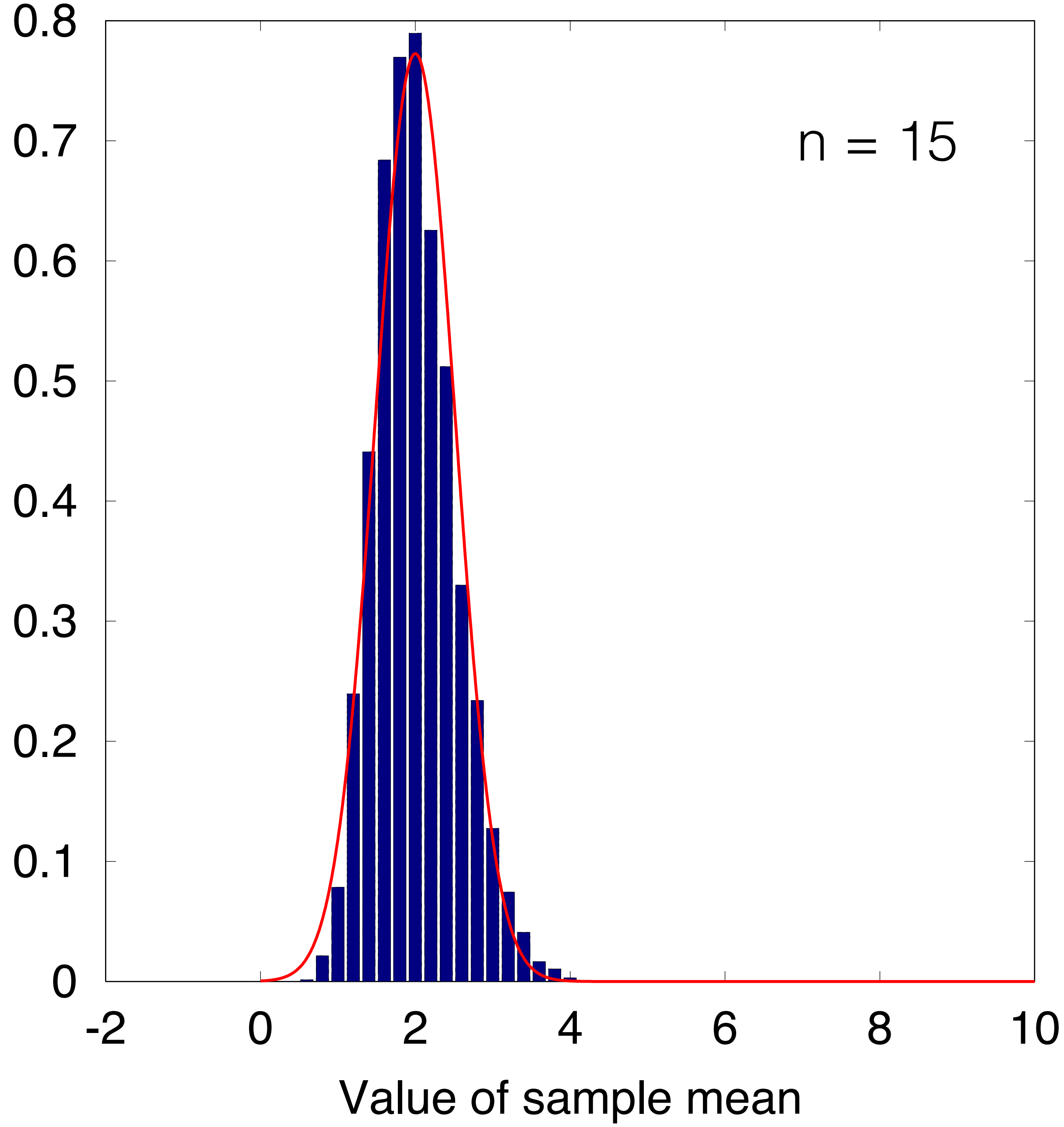
$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad\qquad \overline{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\xrightarrow{\text{if } n \text{ is large enough } (n \geq 30)}$$

$$S_n = \sum_{i=1}^{n} X_i \qquad\qquad S_n \sim \mathcal{N}\left(n\mu, n\sigma^2\right)$$

Distribution of Sample Mean





Distribution of sample mean. Samples are drawn from a population with Exp(0.5)

Example 1

- In a large school, the mean weight of the students is 53.4 kg, and the standard deviation is 6 kg. A random sample of 50 students is drawn. What is the probability that the average weight of these students will be greater than 55 kg?

Confidence Intervals

- Sample mean is only an estimator of the true mean. It is not exactly equal to the true mean.
- We want to assess the level of *confidence* that the true mean falls within a certain interval

Confidence Interval for Population Mean

- Let $X_1, X_2, \ldots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ.

- We know that

$$\overline{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

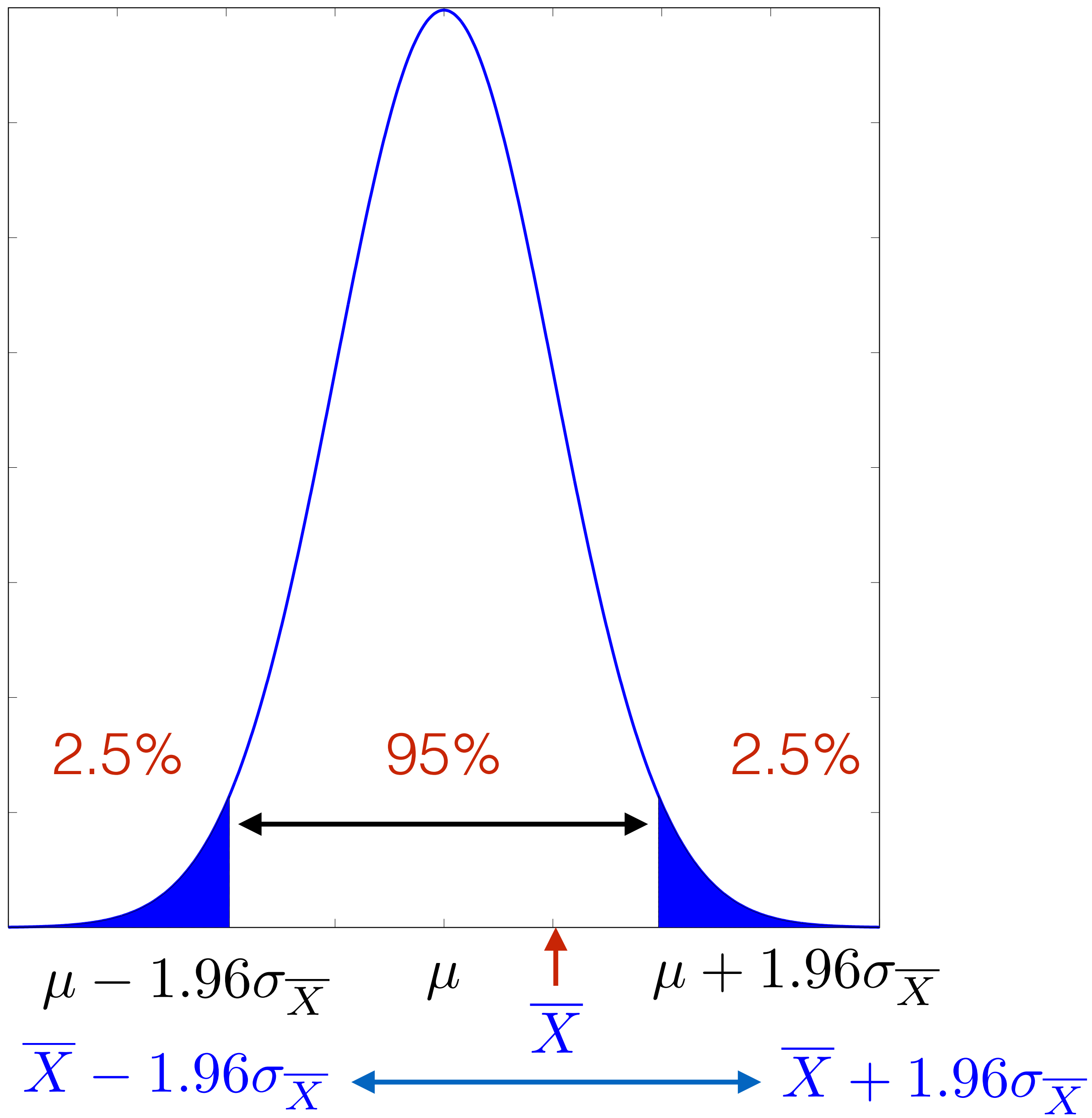
$$\sigma_{\overline{X}} = \sigma/\sqrt{n}$$

- The middle 95% is within

$$\mu \pm 1.96\sigma_{\overline{X}}$$

95% confidence interval

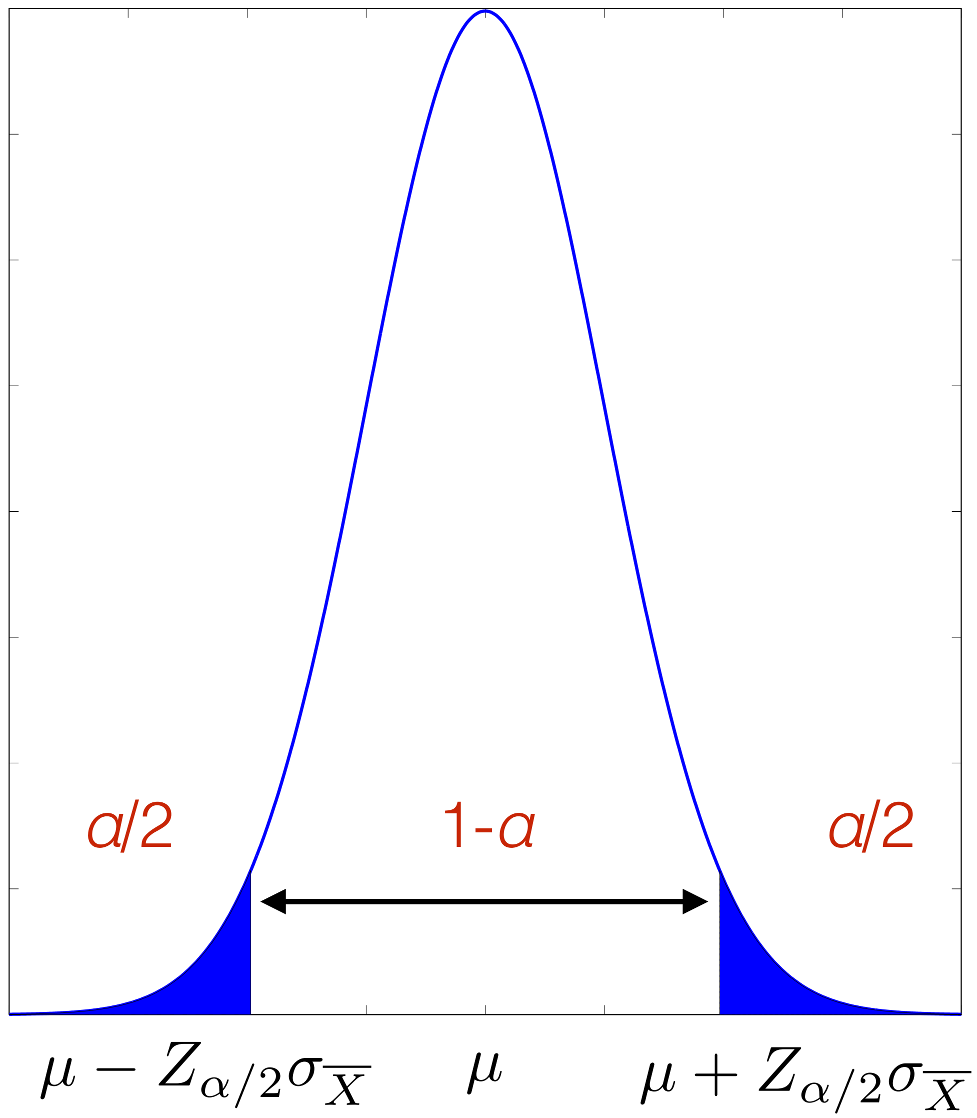
$$\overline{X} \pm 1.96\sigma_{\overline{X}}$$



Confidence Interval

- Let X_1, X_2, …, X_n be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ.
- Let α be a number in [0,1]
- Let $z_{\alpha/2}$ be the z-score that cuts off an area of $\alpha/2$ in the right-tail
- 100(1-α)% confidence interval for μ is

$$\overline{X} \pm Z_{\alpha/2}\sigma_{\overline{X}}$$



Large-Sample Confidence Intervals for a Population Mean

- Let $X_1, X_2, \ldots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ, so that the sample mean $\overline{X}$ is approximately normal. Then a level 100(1- α)% confidence interval for μ is

$$\boxed{\overline{X} \pm Z_{\alpha/2}\, \sigma_{\overline{X}}}$$

where $Z_{\alpha/2}$ is the z-score that cuts off an area of $\alpha/2$ on the right-tail

$$\sigma_{\overline{X}} = \sigma/\sqrt{n}$$

When the value of σ is unknown, it can be replaced with sample standard deviation s.

Values of $Z_{\alpha/2}$ at different C.I.

C.I. Level	$Z_{\alpha/2}$
68%	1
90%	1.645
95%	1.96
99%	2.58
99.7%	3

Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are $\overline{X}$ = 12.05 and s = 0.1. Find
  - 68% confidence interval

Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are $\overline{X}$ = 12.05 and s = 0.1. Find
    - 85% confidence interval

Example 3

- A sample of 50 students has the following sample mean and standard deviation for heights, $\overline{X}$ = 164.2 cm and s = 12.5 cm. Find
  - 68% confidence interval
  - 95% confidence interval
  - 99% confidence interval

Example 4

- Find the levels of the confidence intervals that have the following values of $Z_{\alpha/2}$
  - $Z_{\alpha/2} = 1.28$
  - $Z_{\alpha/2} = 2.17$
  - $Z_{\alpha/2} = 3.28$

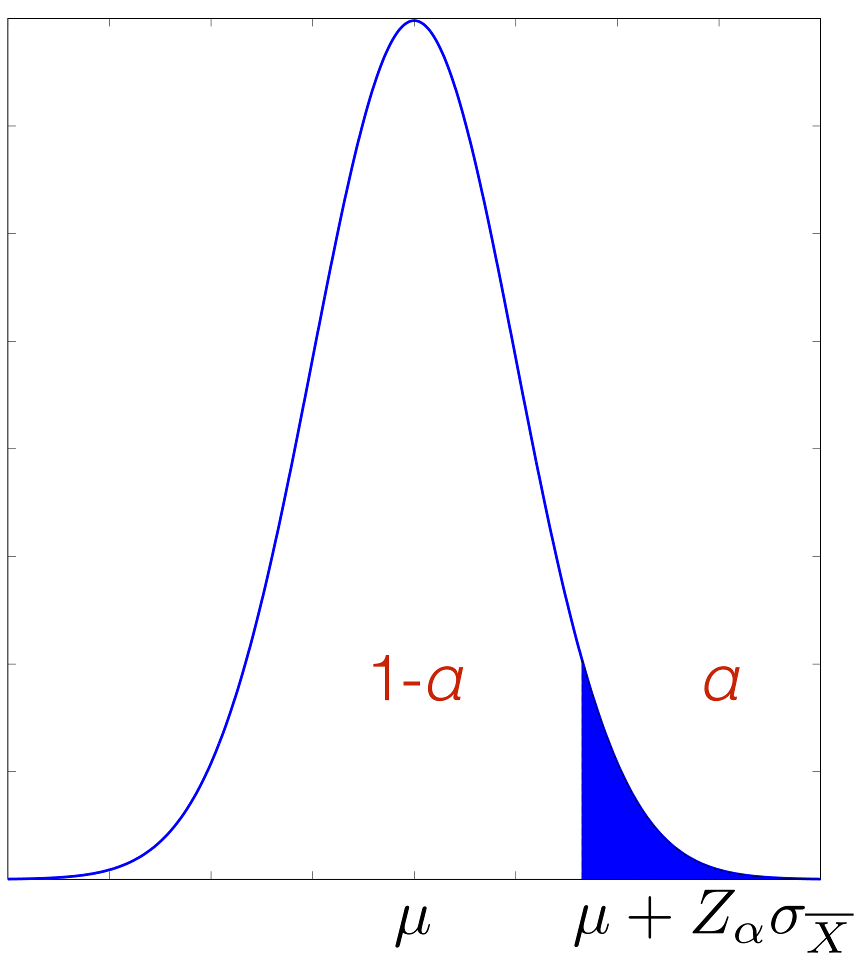
Determining Sample Size Needed for a C.I. with Specified Width

- Sometimes the C.I. is too wide to be useful, and we want to make it narrower. To do this, the sample size must be increased.

- Ex: The sample standard deviation of weights from 100 boxes was s = 0.1 oz. How many boxes must be sampled to obtain a 99% confidence interval of width ±0.012 oz

One-sided C.I.

- The confidence intervals discussed so far have been two-sided in that they specify both a lower and upper confidence bounds
- In some applications, we are only interested only in one of these bounds



One-sided C.I. (2)

- Let $X_1, X_2, \ldots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ, so that the sample mean $\overline{X}$ is approximately normal. Then a level 100(1- α)% lower confidence interval bound for μ is

$$\overline{X} - Z_\alpha\, \sigma_{\overline{X}}$$

and level 100(1- α)% upper confidence bound or μ is

$$\overline{X} + Z_\alpha\, \sigma_{\overline{X}}$$

When the value of σ is unknown, it can be replaced with sample standard deviation *s*.

Values of Z_α at different C.I. (One-sided)

C.I. Level	Z_α
90%	1.28
95%	1.645
99%	2.33

Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights, $\overline{X} = 164.2$ cm and s = 12.5 cm. Find
  - one-sided 95% lower confidence bound

Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights, $\overline{X} = 164.2$ cm and s = 12.5 cm. Find
    - one-sided 99% upper confidence bound